

motives, consumer electronics, aerospace, and many, many more. Four vendors—Autodesk, Dassault Systemes, PTC, and UGS Corp. hold more than half the CAD software market.

Opportunities

Today, the use of CAD/CAM tools has become a necessity in organisations over industry verticals such as automotive, aerospace and utilities. According to a study by Booz Allen Hamilton—a global strategy and technology consulting firm—and NASSCOM, called *Globalization of Engineering services—The Next Growth Frontier for India*, an additional \$40 billion in opportunities will be created from the offshoring of engineering services for the Indian IT Industry by 2020, which implies that the demand for qualified engineers and skilled professionals will rise in the near future.

“With a global engineering shortage, CAD/CAM skills are in great demand. It is important that all students understand the culture of different countries and regions that the companies work in. It might involve a product that is designed in Europe, with programming in Australia, a mould base created in Korea, raw material from Canada, and manufactured in India,” says Ved Narayan, vice president, Asia Pacific Operations, SolidWorks Corporation, who develop and market 3D CAD technology software tools.

In An Organization

The CAD/CAM technologies used across various industries can be broadly divided into Architecture, Construction and Infrastructure, and Manufacturing and Engineering. Different industries such as Automotive and Aerospace come under Manufacturing and Engineering. Today there are a number of super-specialised courses available. By pursuing a super specialised course and having a keen interest in a particular domain, one can look for opportunities in three broad categories:

Designer/Professional

After the education and training acquired from academic institutions, one can work with CAD/CAM tools to design in 2D and 3D, animate models, design simulations of a 3D object simi-



Initiatives of industry-academia collaborative efforts to standardise the level of education and closer exposure to industries will help in meeting the demand of a skilled workforce in the long run”

Rohit Biddappa
Marketing manager
Parametric Technology
(India) Ltd

lar to a desired physical object, and model solids and surfaces. Also, on the construction, architecture, and infrastructure side, one is required to create walkthrough animations to give an idea of nearly exact dimensions of the desired outcomes. Involvement here is more on the design of the product, and in the case of CAM, in deriving codes for Numerical Controls and Computer Numerical Controls.

Consultant

An individual can be a consultant draughtsman, designer, 3D solid modeller, etc. only after experience of at least five to seven years in an industry domain. One needs knowledge of the manufacturing process flow of different industries.

Instructor/Trainer

An instructor or trainer for CAD/CAM tools must have a lot of knowledge and experience of working with those tools, and also industry exposure. Such professionals get hired by academies and training institutes as well as by those involved in organising corporate training sessions.

Prerequisites

The entry ticket for a CAD/CAM job would be an engineering degree. One must have a bachelors or masters degree with specialisation in the mechanical, electrical, or electronics branches. Those with specialised qualification in industrial design are also likely to be recruited across industry verticals. Diploma holders (mechanical, electronics, electrical, industrial design), too, are absorbed.

Candidates with mechanical engineering degrees are preferred since their curriculum allows specialisation in CAD/CAM in the final year. Civil engineers, architects, and computer science graduates, too, can get absorbed in the respective industries as in construction, architectural, and the application programming development domains. For CAE, a post-graduate or doctorate-level degree is necessary.

Engineering education can be pursued from a college that is accredited by the National Board of Accreditation (NBA) under the All India Council for Technical Education (AICTE). In the recent past, issues were raised over educational practices at engineering col-

In A Nutshell

CAD, CAM and CAE—The Difference

The use of computer-based software and hardware tools is made for the design, planning, analysis, manufacturing, and simulation of a product or construction. These tools help in saving time and cost of producing a prototype for testing and analysis, which resulted in its adoption amongst various industry verticals.

Computer Aided Design (CAD) means the use of hardware and software tools by design professionals, engineers, and architects for design, development, modification, and optimisation of products before the

actual manufacturing starts. There are various software by which CAD can be carried out—for example, PRO E, IDEA, CATIA, AutoCAD, and more. These tools are used extensively in the automotive industry, where the design of a model is tested by computer simulation rather than physical prototype testing.

In Computer Aided Manufacturing (CAM), use of specialised computers and CAD tools is made to control, monitor, and adjust the machining tools in a manufacturing unit. A Computer Numerical Control (CNC) machine is programmed after adding raw material to it, and the CNC machine will process

the raw material according to the instructions coded in the programming language (G-code) for Numerical Controls and CNCs.

Computer Aided Engineering (CAE) refers to the use of computer-based analysis tools for optimising engineering tasks at every level of a product's life cycle. The tools are used for engineering tasks such as analysis, design and simulation, planning, diagnosing, validating, and repairing. Tasks such as stress analysis of a component or assembly line, product or process optimisation, mechanical event simulation, and more come under CAE.